



NSW Education Standards Authority

2020 HIGHER SCHOOL CERTIFICATE EXAMINATION

Food Technology

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen

**Total marks:
100****Section I – 20 marks** (pages 2–6)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 50 marks (pages 9–17)

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III – 15 marks (page 33)

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV – 15 marks (page 34)

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** Which sector of the Australian food industry sells packaged frozen pizza?

 - A. Food retail
 - B. Agriculture and fisheries
 - C. Food service and catering
 - D. Food processing/manufacturing

- 2** An ice-cream manufacturer develops a lactose free variety of their most popular ice-cream.

Which type of food product development is this an example of?

 - A. Me too
 - B. Copycat
 - C. New to world
 - D. Line extension

- 3** An elderly person who consumes a low fibre diet is at risk of developing which dietary disorder?

 - A. Anaemia
 - B. Diverticulitis
 - C. Dental caries
 - D. Coeliac disease

- 4 A food product developer creates a single serve, ready-to-heat meal.

Which is the most likely driver to have led to the creation of this food?

- A. Societal changes
- B. Market concerns for health
- C. Consumer demand for value
- D. Market concerns for the environment

- 5 Which of the following is an example of value adding in food manufacturing?

- A. Labelling on packaging
- B. Adding a profit margin to the price of the product
- C. Increasing the price of a product to reflect increased ingredient cost
- D. Packaging a variety of lettuce leaves in a convenient single serve bowl

- 6 In which step of food product development does the packaging design occur?

- A. Feasibility study
- B. Product specifications
- C. Idea generation screening
- D. Testing product prototype

- 7 A food manufacturer is conducting a SWOT analysis after a competitor releases a *me too* of one of its products.

In the SWOT analysis the *me too* product is considered

- A. a threat.
- B. a strength.
- C. a weakness.
- D. an opportunity.

- 8** A company wishes to ensure the consistency and safety of its products throughout production.

An example of a quality management system they might use to achieve this is

- A. CSIRO.
 - B. FSANZ.
 - C. HACCP.
 - D. NHMRC.
- 9** One short-term benefit of using ecologically sustainable production methods is
- A. improved yields.
 - B. reduced labour costs.
 - C. lower production costs.
 - D. reduced carbon emissions.
- 10** Which food additive ensures the free flow of powdered products?
- A. Emulsifier
 - B. Antioxidant
 - C. Bulking agent
 - D. Anti-caking agent
- 11** Which pricing structure involves selling a new product at a similar price to that of existing products?
- A. Break-even
 - B. Competitive
 - C. Penetration
 - D. Skimming

- 12** In which list of career opportunities are all of the positions available in the food processing and manufacturing sector of the Australian food industry?
- A. Fruit picker, food merchandiser, market researcher
 - B. Food technologist, agronomist, warehouse personnel
 - C. Graphic designer, market researcher, food production line worker
 - D. Food technologist, food product developer, product demonstrator
- 13** A government initiative promoting health to Australians is
- A. Dance 4 Diabetes.
 - B. Jump Rope for Heart.
 - C. Australia's Biggest Morning Tea.
 - D. Australian Guide to Healthy Eating.
- 14** How are vegetables processed to deactivate enzymes prior to freezing?
- A. Peeled
 - B. Chopped
 - C. Blanched
 - D. Vacuum sealed
- 15** Raw materials used in food manufacture primarily undergo quality control to ensure that they
- A. minimise waste.
 - B. are cost effective.
 - C. comply to standard specifications.
 - D. comply with work health and safety regulations.

- 16** Which of the following diet-related disorders has the highest economic cost in Australia?
- A. Cardiovascular disease
 - B. Gluten intolerance
 - C. Type 2 diabetes
 - D. Osteoporosis
- 17** Omega 3 oils may be used as a
- A. probiotic.
 - B. flavonoid.
 - C. dietary fibre.
 - D. nutritional supplement.
- 18** In large factories, bulk dry ingredients are stored in a
- A. silo.
 - B. hopper.
 - C. retort.
 - D. centrifuge.
- 19** What is the effect on dried herbs of long-term exposure to light?
- A. Attracts pests
 - B. Discolouration
 - C. Change in texture
 - D. Cultivates microbes
- 20** Australian researchers have recently developed a new variety of plum which can reduce cholesterol levels within 30 minutes.

Which compound in the plum is most likely to deliver this health benefit?

- A. Insoluble fibre
- B. Low-density lipoproteins
- C. Phytochemicals
- D. Vitamin C

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Centre Number

Food Technology

Sections II, III and IV

Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page.
- Use this answer booklet for Sections II, III and IV.
- Follow the instructions inside this booklet as to where you should write your answers.

Please turn over

Section II

50 marks

Attempt Questions 21–26

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (4 marks)

Name an organisation in the Australian food industry.

Name of organisation:

- (a) Identify the sector of the Australian food industry to which this organisation belongs. **1**

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- (b) Describe how this organisation has responded to consumer demand for convenience foods. **3**

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Question 22 (12 marks)

- (a) Outline TWO causes of food spoilage.

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- (b) Explain TWO reasons why manufacturers preserve foods.

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Question 22 continues on page 12

Question 22 (continued)

- (c) Explain the principles behind ONE preservation technique. In your answer, refer to an example of a specific food.

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End of Question 22

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Question 23 (7 marks)

- (a) Define the term *active non-nutrient*.

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- (b) A food manufacturing company has developed a shelf-stable, dairy-free probiotic spread.

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Describe the health benefits of this product for consumers.

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Question 24 (11 marks)

- (a) Outline ONE government policy that affects the Australian food industry. **2**

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- (b) Describe how government legislation influences the decisions made by Australian food manufacturers. **3**

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- (c) Explain external factors that affect food product development in the Australian food industry. **6**

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Question 25 (10 marks)

- (a) Identify the types of market research that would be useful during the development of a food product. **2**

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- (b) Why is it important to test product prototypes in food product development? **3**

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Question 25 continues on page 16

Question 25 (continued)

- (c) Explain the relationship between nutrient intake and ONE dietary disorder.

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Name of disorder:

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End of Question 25

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Question 26 (6 marks)

How is quality management in food manufacture assisted by computerised production systems?

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Section III — Answer Question 27 on pages 18–24

27

Question Number

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If you require more space to answer Section III, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section III, tick here.

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Section IV — Answer Question 28 on pages 25–31

28

Question Number

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If you require more space to answer Section IV, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section IV, tick here.

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Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question on pages 18–24 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Question 27 (15 marks)

A food manufacturing company has produced a plant-based product that is low in cholesterol and high in fibre and protein.

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|--|----------|
| (a) Outline a target market for this product. | 2 |
| (b) Propose a marketing plan that could be used to launch this product. | 5 |
| (c) Explain the potential benefits of moving towards a plant-based diet. | 8 |

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question on pages 25–31 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
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Question 28 (15 marks)

How effectively have developments in food packaging responded to both consumer demands for convenience and concerns about the environment?

End of paper